SolarEdge

Solar Inverter

Complete Installation Manual

Chapter 1: Introducing the SolarEdge System

System Overview

The SolarEdge power harvesting solution maximizes the power output from any type of solar Photovoltaic (PV) installation while reducing the average cost per watt. The system consists of three main components.

SolarEdge Power Optimizer

SolarEdge power optimizers are DC-DC converters connected to PV modules in order to maximize power harvesting by performing independent Maximum Power Point Tracking (MPPT) at the module level.

Key characteristics:

- Regulate string voltage at a constant level, regardless of string length and environmental conditions.
- Include a safety voltage function that automatically reduces output to 1 Vdc during fault conditions, when disconnected from the inverter, or when the inverter's ON/OFF switch is turned OFF.
- Transmit module performance data over the DC power line to the inverter.

Two types of power optimizers are available:

- Module Add-on power optimizer – connected to one or more modules
- Module embedded power optimizer – embedded into a module

SolarEdge Inverter

The SolarEdge inverter efficiently converts DC power from the modules into AC power that can be fed into the main AC service of the site and from there to the grid. The inverter also receives the monitoring data from each power optimizer and transmits it to the SolarEdge monitoring portal (requires Internet connection).

SolarEdge Monitoring Portal

The SolarEdge monitoring portal enables you to monitor the technical and financial performance of one or more SolarEdge sites. It provides information about present and past performance of each module individually and about the system as a whole.

Installation Equipment List

Standard tools needed for installation:

- Allen screwdriver for M6/M8 screw types
- Standard flat head screwdrivers set
- Voltage tester screwdriver
- Drilling motor and bits suitable for the surface
- M6 (1/4") or M8 (5/16") stainless bolts, nuts, and washers for power optimizer racking
- MC4 Crimper
- Wire cutters and wire strippers
- Voltmeter

For communication options, additionally:

- Ethernet: CAT5/6 twisted pair cable, RJ-45 plugs, RJ-45 crimper
- RS485: Four- or six-wire twisted pair, small flat head screwdriver

Inverter Transport and Storage

Transport the inverter in its original packaging, facing up and without exposing it to unnecessary shocks. If the original package is unavailable, use a similar box that can withstand the inverter weight, has a handle system, and can be closed fully.

NOTE: Store the inverter in a dry place where ambient temperatures are -25°C to +65°C (-13°F to 149°F).

Chapter 2: Handling and Safety Instructions

During installation, testing and inspection, adherence to all handling and safety instructions is mandatory.

Safety Symbols

WARNING!	Denotes a hazard that, if not correctly performed or adhered to, could result in injury or loss of life.
CAUTION:	Denotes a hazard that could result in damage or destruction of the product.
NOTE:	Denotes additional information about the current subject.
IMPORTANT SAFETY FEATURE:	Denotes information about safety issues.

Safety Instructions

WARNING: The cover must be opened only after shutting off the inverter ON/OFF switch located at the bottom of the inverter. This disables the DC voltage inside the inverter. Wait five minutes before opening the cover. Otherwise, there is a risk of electric shock from energy stored in the capacitors.

WARNING: Before operating the inverter, ensure that the AC power cable and wall outlet are grounded properly.

WARNING: Opening the inverter and repairing or testing under power must be performed only by qualified service personnel familiar with this inverter.

WARNING: Do not touch the PV panels or any rail system connected when the inverter switch is ON, unless grounded.

CAUTION: This unit must be operated under the specified operating specifications, as described in the latest technical specification datasheet, available on the SolarEdge website.

CAUTION: The inverters are IP65 rated. Unused connectors and glands should be sealed with the provided seals.

NOTE: Use PV modules rated according to IEC 61730 class A.

Warning Symbols on Inverter Label

Lightning bolt	Risk of electric shock
Lightning bolt + 5 minutes	Risk of electric shock from energy stored in capacitor. Do not remove cover until 5 minutes after disconnecting all sources of supply.
Heat waves / flame	Hot surface – To reduce the risk of burns, do not touch.
ON/OFF switch: 0=OFF, 1=ON	Inverter power switch position indicator

Chapter 3: Installing the Power Optimizers

Safety Notes

WARNING: When modifying an existing installation, turn OFF the inverter ON/OFF switch and the AC circuit breaker on the main AC distribution panel.

WARNING: Input and output connectors are not watertight until mated. Open connectors should be mated to each other or plugged with appropriate watertight caps.

CAUTION: Cutting the power optimizer input or output cable connector is prohibited and will void the warranty.

CAUTION: Power optimizers are IP65/NEMA4 rated. Choose a mounting location where optimizers will not be submerged in water.

CAUTION: If you intend to mount the optimizers directly to the module or module frame, first consult the module manufacturer for guidance regarding the mounting location.

CAUTION: Installing a SolarEdge system without ensuring compatibility of the module connectors with the optimizer connectors may be unsafe and could cause functionality problems such as ground faults.

NOTE: IMPORTANT SAFETY FEATURE: Modules with SolarEdge power optimizers carry only a low safety voltage (1V) before the inverter is turned ON. As long as power optimizers are not connected to the inverter or the inverter is OFF, each optimizer will output a safe voltage of 1V.

Installation Guidelines

- This chapter refers only to module add-on power optimizers. For module-embedded models, refer to the installation instructions supplied with the embedded module.
- The power optimizer can be placed in any orientation.
- Position the power optimizer close enough to its module so that their cables can be connected.
- To allow proper heat dissipation, maintain 1in / 2.5 cm clearance between the power optimizer and other surfaces.
- Minimum and maximum string length guidelines are stated in the power optimizer datasheets.
- Completely shaded modules may cause their power optimizers to temporarily shut down, but will not affect other optimizers in the string.
- Do not leave power optimizer connectors uncovered. Open connectors should be mated to each other.

Step 1: Mounting the Power Optimizers

1. Determine the power optimizer mounting location and use the mounting brackets to attach it to the support structure.
2. If required, mark the mounting hole locations and drill the holes.

CAUTION: Do not drill through the power optimizer or through the mounting holes. Drilling vibrations can damage the power optimizer and will void the warranty.

3. Attach each power optimizer to the rack using M6 (1/4") or M8 (5/16") stainless steel bolts, nuts, and washers.
4. Verify that each power optimizer is securely attached to the supporting structure or module.

NOTE: Record power optimizer serial numbers and locations for monitoring purposes.

Step 2: Connecting a Module to a Power Optimizer

- Connect the Plus (+) output connector of the module to the Plus (+) input connector of the power optimizer.
- Connect the Minus (-) output connector of the module to the Minus (-) input connector of the power optimizer.

NOTE: Images are for illustration purposes only. Refer to the label on the product to identify the plus and minus input and output connectors.

Step 3: Connecting Power Optimizers in Strings

You can construct parallel strings of unequal length. The number of power optimizers in each string does not have to be the same. The minimum and maximum string lengths are specified in the power optimizers datasheets.

NOTE: String length should not exceed 300m from DC+ to DC- of the inverter. Use at least 4 mm² DC cables.

5. Connect the Minus (-) output connector of the string's first power optimizer to the Plus (+) output connector of the string's second power optimizer.
6. Connect the rest of the power optimizers in the string in the same manner.
7. If you intend to monitor the installation, record the physical location of each power optimizer.

CAUTION: Do not leave the power optimizer connectors disconnected. Open connectors should be mated to each other.

Step 4: Verifying Proper Power Optimizer Connection

After a module is connected to a power optimizer, the power optimizer outputs a safe voltage of 1V. The total string voltage should equal approximately 1V times the number of power optimizers connected in series.

Example: If 10 power optimizers are connected in a string, 10V should be produced.

- Make sure the modules are exposed to sunlight during this process; otherwise, the power optimizers may not be powered.
- If using a tracker, the power optimizer will turn ON only if the tracker is tracking the sun and the module provides at least 2W.
- Isolation measurement using a device of up to 1000V is acceptable.

8. Measure the voltage of each string individually before connecting it to other strings or to the inverter. Verify correct polarity using a voltmeter with at least 0.1V measurement accuracy.
9. For troubleshooting, refer to Power Optimizers Troubleshooting in Chapter 7.

Chapter 4: Installing the Inverter

Install the inverter either before or after the modules and power optimizers have been installed.

> **CAUTION:** Do not rest the connectors at the bottom of the inverter on the ground, as it may damage them. To rest the inverter on the ground, lay it on its back, front, or side.

Inverter Package Contents

- One SolarEdge inverter
- One mounting bracket
- Two Allen screws for fastening the inverter to the mounting bracket
- This installation guide
- Quick installation guide
- Inverter activation instructions (may include an activation card)

Identifying the Inverter

Refer to the sticker on the inverter that specifies its Serial Number and Electrical Ratings. Provide the serial number when contacting SolarEdge support. The serial number is also required when opening a new site in the SolarEdge monitoring portal.

Inverter Interfaces

All connectors and components are located at the bottom of the inverter:

LCD panel	Displays inverter information and configuration parameters
LCD LEDs (Green, Yellow, Red)	Indicate inverter status (see LED table below)
ON/OFF switch	Starts/stops power optimizer operation and power production
LCD light button	Lights up LCD for 30 seconds; also accesses configuration menus
Communication glands	Two glands for communication options (3 openings each)
DC inputs	Connection points for the PV installation strings
AC output	Grid connection; single-phase: PG21 (9-16mm); three-phase: M32 (15-21mm)

LED Status Indicators

Color	Description	ON	Blinking	OFF
Green	Power production	Producing power	Standby mode – waiting for working voltage	Not producing (night/switch OFF/error)
Yellow	Module comms / shutdown	—	Receiving optimizer data OR inverter shutting down	Normal operation
Red	Fault	Error present – see Error Codes	Inverter shutting down	No fault
All LEDs	Configuration	All ON during Setup mode		

CAUTION: Do not remove the six screws on the DC metal panel as it may harm the inverter sealing and void the warranty.

Mounting the Inverter

Clearance Requirements

- 8in / 20 cm to the top and bottom of the inverter
- 4in / 10 cm to the right and left of the inverter
- 16in / 40 cm between inverters installed side by side
- 16in / 40 cm between inverters installed one above the other

Mounting Steps

10. Position the mounting bracket against the wall/pole. Ensure the U-shaped indentations face up and the 'UP' marking is correctly oriented.
11. Use at least two bracket holes. Determine which holes to use based on mounting surface type and material.

NOTE: The mounting surface must support the inverter weight: single phase 20-21 kg; three phase 33 kg. The bracket weighs 1.3 kg.

12. Drill the holes and mount the bracket. Verify the bracket is firmly attached.
13. Hang the inverter on the bracket using the screws at the top. Let the inverter lay flat against the wall or pole.
14. Insert the two supplied screws through the outer heat sink fin on both sides into the bracket. Tighten to 4.0 N*m / 2.9 lb*ft.

CAUTION: HEAVY OBJECT. Single phase inverter: 20-21 kg. Three phase: 33 kg. Use proper lifting techniques and a lifting aid if required.

Connecting the AC to the Inverter

Grid Connection Guidelines

NOTE: Three phase inverters require neutral connection at all times (only grids with neutral connection are supported).

- For installation on a delta grid without neutral, multiple single phase inverters can be used.
- When connecting multiple single phase inverters to a three-phase grid, phase balancing may be required.

AC Wiring Steps

Use a three-wire cable for single phase or five-wire cable for three phase. Maximum wire size: 16 mm².

15. Turn OFF the AC circuit breaker.
16. Open the inverter cover's six Allen screws and carefully move the cover vertically before lowering it.

CAUTION: When removing the cover, make sure not to damage internal components.

17. Strip 58 mm / 2.32" of the external cable insulation and strip 8 mm / 0.32" of the internal wire insulation.
18. Open the AC cable gland and insert the cable through the gland.

WARNING: Turn OFF the AC before connecting the AC terminals. Connect the equipment grounding wire FIRST, before connecting the AC Line and Neutral wires.

Single Phase AC Terminal Connections

Protective Earth	PE terminal
Line	L terminal
Neutral	N terminal

Three Phase AC Terminal Connections

Line 1	L1 terminal
Line 2	L2 terminal
Line 3	L3 terminal (feeds the internal power supply)
Protective Earth	PE terminal
Neutral	N terminal

NOTE: If power control is enabled, respect the order of grid lines. A 120-degree phase difference must be maintained: L1-L2-L3 order. If wrong order, an error is displayed on LCD and the inverter will not produce power.

19. Tighten screws of each terminal with torque 1.2–1.5 N*m / 0.88–1.1 lb*ft.
20. Check that the wires are fully inserted and cannot be pulled out easily.
21. Tighten the AC cable gland with torque 2.8–3.3 N*m / 2.0–2.4 lb*ft.
22. Verify there are no unconnected wires and unused terminal screws are tightened.

Connecting the Strings to the Inverter

Connect the string to the DC input pairs. If required, connect additional strings in parallel using an external combiner box/branch cables before connecting to the inverter.

NOTE: Functional electrical earthing of DC-side negative or positive poles is prohibited because the inverter has no transformer. Grounding of module frames and mounting equipment is acceptable.

NOTE: SolarEdge's fixed input voltage architecture enables parallel strings of different lengths. They do not need to have the same number of power optimizers, as long as each string length is within the permitted range.

23. Connect the DC connectors of each string to the DC+ and DC- connectors according to the labels on the inverter.

Selecting a Residual-Current Device (RCD)

All SolarEdge inverters incorporate a certified internal Residual Current Device (RCD). There are 2 trip thresholds: 30 mA (electrocution protection) and 300 mA (slow rising current).

If an external RCD is required by local regulations, SolarEdge recommends a Type-A RCD at 100 mA or 300 mA unless a lower value is required. Type-B RCD may be used when required by local regulations.

Chapter 5: Commissioning the Installation

Step 1: Activating the System

24. Verify that the inverter ON/OFF switch is OFF. Turn ON the AC breaker.

WARNING: ELECTRICAL SHOCK HAZARD. Do not touch uninsulated wires when the inverter cover is removed.

25. Activate the inverter according to the activation instructions supplied in the inverter package.
26. Verify that the inverter is configured to the proper country by pressing the LCD light button until reaching the ID status screen.

NOTE: If an inverter is not configured to any country, it will not produce energy, and 'No Country Selected' is displayed.

27. If required, perform country settings or inverter configuration using the internal LCD user buttons, or configure communication options.
28. Attach the inverter cover and tighten screws with torque 4.0 N*m / 2.9 lb*ft. Tighten corner screws first, then the two central screws.
29. If an additional external DC switch is installed between the power optimizers and the inverter, turn it ON.

Initial LCD Status Screen Fields

P_OK	Connection with power optimizers – appears on first telemetry reception
000/000	Number of power optimizers paired (at this stage: 000 since none paired yet)
S_OK	Connection to SolarEdge monitoring portal is successful
Vac [V]	Grid AC output voltage – verify correct value
Vdc [V]	DC input voltage – safety voltage of 1V per optimizer in the string
Pac [W]	AC output power – should be 0.0 since the inverter is OFF
OFF	Inverter ON/OFF switch is in the OFF position

NOTE: A measurement error on the inverter LCD of ±3V is acceptable.

Step 2: Pairing Power Optimizers to the Inverter

All power optimizers must be logically paired to their inverter. Optimizers do not start producing power until paired. Perform this step when modules are exposed to sunlight.

30. Verify that the inverter ON/OFF switch is OFF.
31. Press and hold down the inverter LCD light button for about 10 seconds until the pairing message is displayed.
32. Keep holding for 5 seconds until the pairing screen appears showing 'Turn Switch To On'.
33. Turn the inverter ON/OFF switch to ON within 5 seconds. If you wait longer, the inverter exits pairing mode.
34. Wait for the completion of the pairing (remaining seconds reaches 0).

When pairing succeeds, 'Pairing Completed' is displayed.

If pairing fails, an error is displayed. Repeat the pairing steps. If the problem persists, contact SolarEdge Support.

Step 3: Verifying Proper Operation

After the wake-up time, verify the following on the LCD panel:

- The green inverter LED is steadily lit.
- The ON/OFF indicator on the LCD panel reads ON.
- P_OK: XXX/YYY appears, indicating connection to power optimizers.

NOTE: Optimizers send telemetries in a frequency of up to 10 minutes. It can take up to 20 minutes for all to be counted on the LCD screen.

- S_OK appears if the inverter is connected to the SolarEdge monitoring portal.
- Vac [V]: Measured grid AC output voltage.
- Vdc [V]: DC input voltage – 250V to 450V (single phase) or 550V to 950V (three phase).
- Pac [W]: Total AC output power produced.

Take note of the serial number on the inverter label. This is needed to open a new site in the monitoring portal.

Step 4: Reporting and Monitoring Installation Data

NOTE: This step requires connecting one of the communication options (see Chapter 6).

Providing Installation Information

Two methods are available to map installed power optimizers:

- Paper Template: Fill out the Physical Layout Template using the detachable 2D barcode stickers on each power optimizer. Scan and upload to the SolarEdge monitoring portal during site registration.
- iPhone Site Mapper: Use the SolarEdge Site Mapper iPhone application to scan the 2D barcodes. The app creates an XML file to upload during site registration.

Create the site in the monitoring portal at: http://www.solaredge.com/groups/site-registration

Chapter 6: User Interface

LCD User Buttons

Four buttons are located inside the inverter, above the LCD panel:

Esc	Moves cursor to beginning of current parameter; goes to previous menu; cancels a value change with long press (until 'Aborted' is displayed)
Up (1)	Moves cursor between menu options; moves between characters of a parameter; toggles between possible characters when setting a value
Down (2)	Same as Up but in reverse direction
Enter (3)	Selects a menu option; accepts a value change with long press (until 'Applied' is displayed)

NOTE: Use the three rightmost buttons for entering '123' when entering the password.

Inverter Configuration – Setup Mode

You can configure the inverter using one of two methods:

- Internal LCD user buttons (cover removed) – full configuration menu available.
- External LCD light button (cover NOT required) – fewer menus but same core functionality.

Configuring Using Internal LCD User Buttons

35. Turn the inverter ON/OFF switch to OFF (AC remains ON).

WARNING: If the inverter is operating, 'DC VOLTAGE NOT SAFE – DO NOT DISCONNECT' is displayed until the DC voltage is safe (50V). Do not open cover until voltage is safe or at least 5 minutes have passed.

36. Remove the inverter cover.

WARNING: ELECTRICAL SHOCK HAZARD. Do not touch uninsulated wires when the inverter cover is removed.

37. Press the Enter button for at least 5 seconds. The password prompt is displayed.
38. Use the three right-most buttons to type password: 12312312

The inverter is now in Setup mode and all LEDs are lit. The inverter automatically exits Setup mode if no buttons are pressed for more than 2 minutes.

Configuration Menu Options

Country	Specify the country for grid code compliance. CRITICAL – inverter must be configured to proper country to comply with grid codes.
Language	English, German, Spanish, French, Italian
Communication	Configure server connection method and communication ports (LAN, RS485, ZigBee, RS232)
Power Control	Grid Control, RRCR, Reactive Power, Active Power, Wakeup, Phase Balance, P(f), Advanced, Load Defaults
Display	Temperature units (C/F); LCD On Time (10-120 sec); TLM On Time (1-120 min)
Maintenance	Date and Time; Reset Counters; Factory Reset; SW Upgrade

	SD-Card
Information	Versions (ID, DSP1/2, CPU); Error Log (last 5 errors); Warning Log (last 5 warnings)

Inverter Status Screens – Operational Mode

Pressing the external LCD light button turns on the backlight. Additional presses cycle through the following screens:

1. Initial / Main Inverter Status

Vac [V]	AC output voltage
Vdc [V]	DC input voltage
Pac [W]	AC output power
Fac [Hz]	AC output frequency
OPs_Ok	Number of optimizers sending telemetries (paired)
Temp [C or F]	Inverter heat sink temperature
P_OK: XXX/YYY	XXX = telemetries received last 2 hours; YYY = total paired optimizers
S_OK	Connection to SolarEdge monitoring portal is successful
ON/OFF	Position of the inverter ON/OFF switch

2. Energy Meter Status

Displays total energy produced during:

Day [Wh]	Energy produced today
Month [KWh]	Energy produced this month
Year [KWh]	Energy produced this year
Total [KWh]	Total energy since inverter installation

NOTE: This data is accumulated according to an internal real-time clock.

3. Telemetry Status

Displays last power optimizer telemetry received. Updates as each optimizer reports.

Module#	Power optimizer serial number
Energy [Wh]	Power optimizer energy output
Vdc_O [V]	Power optimizer output voltage
Vdc_I [V]	Power optimizer input voltage (module voltage)

4. ID Status

Displays inverter software version and country configuration.

ID	Inverter serial number / ID
DSP1/2	DSP digital control board firmware version
CPU	Communication board firmware version
Country	Country to which the inverter is configured

5. Server Communication Status

Server	Method of connection to SolarEdge monitoring portal
S_OK	Connection to monitoring portal is successful
Status	OK if successful connection with specified server port/device
xxxxxxxx	Eight-bit Ethernet communication status (1=OK, 0=error)
Error message	Specific error details if communication fails

6. IP Status

Describes the Ethernet configuration:

IP	Inverter IP address
MSK	Subnet mask
GW	Gateway address
MAC	Media Access Control address

7. ZigBee Status

PAN ID	ZigBee transceiver PAN ID
Ch.	ZigBee transceiver channel
ID	ZigBee transceiver ID
MID	Master ID of the coordinator ZigBee module
RSSI	Receive signal strength: L=Low, M=Medium, H=High, (-)=No signal

8. Fan Health Status (Three Phase Inverters Only)

Operating	Fan is functioning correctly
Not Working	May be a system error, not necessarily a faulty fan. Turn AC off and on to clear.
Replace	Faulty fan – requires replacement. Display blinks constantly until replaced.

9. Power Control Status (if enabled)

P Control	Country according to which power control is done
Peak Power	Ratio of current output to maximum AC power (shows limit if active power is limited)
PF	Power Factor – CosPhi ratio between active and reactive power
AC Power	AC power currently being produced

Chapter 7: Setting Up Communication

Power optimizers send information to the inverter via the DC power lines. No additional wires are required. The inverter then sends data to the SolarEdge monitoring portal through the Internet. Communication setup is not required for power harvesting – only for using the monitoring portal.

> **CAUTION:** When connecting communication cables, the ON/OFF switch and AC must both be turned OFF. When configuring communication parameters, the ON/OFF switch must be OFF but AC must be ON.

Communication Connectors

Gland #	Opening	Functionality	Cable Size (diameter)
1 (PG16)	One small	External antenna cable	2–4 mm
1 (PG16)	Two large	Ethernet (CAT5/6) or RS232	4.5–7 mm
2 (PG13.5)	All three	RS485, power reduction	2.5–5 mm

Communication Types

Ethernet (LAN)	Used for LAN connection. Most straightforward setup for monitoring portal.
RS485 Bus	Most commonly used. Connects multiple SolarEdge devices on same bus (master-slave). Up to 31 slave inverters + 1 master. Also used for external devices (revenue meters, data loggers).
RS232 (UART)	Enables connection to an external SolarEdge GSM modem.
ZigBee	Optional wireless communication (purchased separately).
USB (Mini-B 5-pin)	Enables connection of a laptop to the inverter for configuration.

Removing the Inverter Cover

39. Turn the inverter ON/OFF switch to OFF.
40. Disconnect the AC by turning OFF circuit breakers on distribution panel. Wait 5 minutes for capacitors to discharge.
41. Open the inverter cover's six Allen screws and carefully pull the cover vertically before lowering it.

> **CAUTION:** When removing the cover, make sure not to damage internal components.

Creating an Ethernet (LAN) Connection

This option connects the inverter to the monitoring portal through a LAN. Each inverter sends its monitored data independently to the portal.

Firewall note: Configure to enable connection to prod.solaredge.com on TCP Port 22222 (incoming and outgoing).

42. Remove the inverter cover.
43. Open gland #1.
44. Insert an Ethernet CAT5/6 cable through the opening with rubber waterproof fitting.
45. Depending on inverter connector type (RJ45 or 8-pin terminal block), connect the wires according to the T568B or T568A standard.

46. Connect the other end to the RJ45 port of the Ethernet switch or router.
47. Verify the connection (S_OK should appear on LCD after up to 2 minutes).

Ethernet Wiring Reference (T568B Standard)

RJ45 Pin	Wire Color (T568B)	Wire Color (T568A)	Signal
1	White/Orange	White/Green	Transmit+
2	Orange	Green	Transmit-
3	White/Green	White/Orange	Receive+
4	Blue	Blue	Reserved
5	White/Blue	White/Blue	Reserved
6	Green	Orange	Receive-
7	White/Brown	White/Brown	Reserved
8	Brown	Brown	Reserved

Creating an RS485 Bus Connection

The RS485 option enables creating a bus of up to 31 slave inverters and 1 master inverter connected in a chain.

48. Remove the inverter cover.
49. Remove the seal from an opening in communication gland #2 and insert the wire.
50. Pull out the 9-pin RS485/RS232 terminal block connector.
51. Loosen screws of pins A(+), B(-), and G on the RS-485 terminal block.
52. Insert wire ends into G, A, and B pins. Use four- or six-wire twisted pair cable. Use consistent wire colors across all A, B, and G pins.
53. Connect all B, A, and G pins in all inverters in the chain.

NOTE: Do not cross-connect B, A, and G wires. Do not insert wires into RS485-2 pins.

54. Tighten the terminal block screws.
55. Push the RS485 terminal block firmly into the connector.
56. Terminate the FIRST and LAST device in the chain by switching the termination DIP-switch SW7 to ON (move switch up).

NOTE: Only the first and last SolarEdge devices in the chain should be terminated. Other inverters should have the termination switch OFF.

57. Designate one inverter as the master. Connect the master to the monitoring portal via LAN or GSM modem.

To configure: Select RS485-1 Conf. > Device Type: SolarEdge > Protocol: Master > Slave Detect

Verifying the Connection

58. Close the inverter cover.
59. Turn ON the AC by turning ON the circuit breaker.
60. Wait up to 2 minutes for the inverter to connect to the SolarEdge monitoring portal.

S_OK on the LCD indicates a successful connection to the SolarEdge monitoring portal.

Chapter 8: Errors and Troubleshooting

Error Codes Reference

Error messages include an error number and a description. An error message is displayed for 30 seconds, after which the inverter waits the necessary reconnection time and then restarts.

WARNING: All Warnings and Cautions in this manual should be adhered to while troubleshooting an error.

Error # (1-Ph)	Error # (3-Ph)	LCD Message	Description	Troubleshooting
N/A	1–4	Fan # Failure	One of the fans is not working	Check Fan Health Status screen. Replace fan if needed (replacement kit from SolarEdge).
4,5,8,12,18-23,39,42,45	45,48,50-53,94,108-113	SW Error	Internal software error	If fault persists, contact SolarEdge support.
N/A	112	Wrong AC Connection	Grid lines connected in wrong order (no 120 deg between phases)	Switch L1/L2 or L2/L3 connections.
9, 13	N/A	AC Current Surge	High AC output currents detected – may be due to AC voltage changes or switching loads	Check AC connection; check with grid operator for surge sources.
10, 37, 38	76, 77, 90	Ground Current – RCD	Ground current surge – insufficient insulation to ground	Turn OFF, wait 5 min, disconnect DC strings, test each string separately. Only certified installer should fix.
14	58/59/60	AC Voltage Too High (L1/2/3)	Sudden high AC voltage detected	Check AC connection; verify country setting; use larger gauge wire; consult grid operator.
15	102	DC Voltage Too High	Input DC voltage exceeds maximum supported level	Inverter shuts off optimizers and restarts. If persistent: turn OFF, check strings, re-commission inverters.
16	123	Measurement Error	Internal hardware error	If fault persists, contact SolarEdge support.
17	104	Temperature Too High	Over-temperature condition	Verify clearances; clean heat-sink fins; reinstall in cooler location.
24	N/A	Faulty Temp. Sensor	Broken or unconnected temperature sensor	If fault persists, contact SolarEdge support.
25	121	Isolation Fault	PV array not properly isolated from ground earth	Check PV installation for isolation problems. Only certified installer may fix.
26	122	Faulty AC Relay	AC relay failed during wake-up tests	Disconnect from AC grid; contact SolarEdge support.
29-30	N/A	Phase Balance Error	Balance of each phase (L1-N, L2-N) exceeded allowed limit	Check grid connection, GND wire, L1/L2/Neutral connections; verify symmetric load.
31, 33	64/65/66	AC Voltage Too High	Grid voltage above country limit	Verify country setting; turn OFF inverters; verify AC grid voltage; consult grid

Error # (1-Ph)	Error # (3-Ph)	LCD Message	Description	Troubleshooting
				operator.
32, 41	61-63, 67-69	AC Voltage Too Low	Grid voltage below country limit	Verify country setting; consult grid operator.
34	79/80/81	AC Freq Too High	Grid frequency above country limit	Same as AC Voltage Too Low.
35	82/83/84	AC Freq Too Low	Grid frequency below country limit	Same as AC Voltage Too Low.
36	72/74/75	DC Injection (L1/2/3)	DC feed detected on AC output	If fault persists, contact SolarEdge support.
40	N/A	Islanding	AC grid voltage out of range; inverter shut off due to islanding	Inverter restarts after reconnection time. Consult grid operator if AC disruptions persist.
44	44	No Country Selected	Inverter not configured to any country	Select the country via LCD menu.
46	N/A	Phase Unbalance	Uneven power supply across three inverters	Change Phase Balance option in LCD menu to Disable.
N/A	103, 146	DC Voltage Too Low	Input DC voltage below minimum supported level	Turn inverter OFF then ON. If persists, contact SolarEdge support.
N/A	78	Grid Sync Error	Grid voltage or frequency is unstable	If fault persists, contact SolarEdge support.
N/A	105	Temperature Too Low	Under-temperature condition	Reinstall inverter in a warmer location.
N/A	124	Ground Current – RCD	Internal hardware error	If fault persists, contact SolarEdge support.

Troubleshooting Communication – S_OK Not Displayed

Ethernet Communication

The Server Communication Status window shows an eight-bit status string. 1 = OK, 0 = error.

Bit	Error Message	Description	Troubleshooting
1st	LAN Disconnected	Ethernet cable not connected properly	Check cable pin-out and connection.
2nd	DHCP Failed	Failed to get IP from DHCP server	Check router and inverter configuration.
3rd	Gateway Ping Failed	Connection to router not available	Check physical connection to switch/router.
4th	G Server Ping Failed	Internet not available (google.com ping failed)	Connect laptop and check internet.
5th-7th	Server x Ping Failed	Ping to redundant server failed	Check prod.solaredge.com and port 22222.
8th	Tcp Connect. Failed	Connection to SolarEdge server failed	Check firewall; ensure port 22222 is open.

RS485 Communication

- If slaves reported in master is less than actual slaves: check Server Communication status on each slave inverter (Status: OK should appear).
- If 'Master Not Found' appears: master device is not responsive or RS485 cable is not connected.

General Troubleshooting Steps

61. Check that the modem or hub/router is functioning properly.
62. Check that the connection to the internal connector on the communication board is properly done.
63. Check that the selected communication option is properly configured.
64. Use a method independent of the SolarEdge inverter to check whether the network and modem are operating properly (e.g., connect a laptop to the Ethernet router).
65. Check whether a firewall or another type of network filter is blocking communication.
66. For ZigBee troubleshooting, refer to the ZigBee installation manuals.

Power Optimizers Troubleshooting

Problem	Possible Cause	Possible Solution
String voltage is 0V	Output of one or more optimizers is disconnected	Connect all power optimizer outputs.
String voltage lower than expected	One or more optimizers not connected in string	Connect all power optimizers.
String voltage lower than expected	Modules not connected properly to optimizer inputs	Connect modules to optimizer inputs.
String voltage lower than expected	Wrong polarity connection on one of the strings	Check polarity with voltmeter and correct.
String voltage higher than expected	Extra power optimizer connected in string	Check if extra optimizer is in the string.
String voltage higher than expected	Module connected directly to string without optimizer	Verify only optimizers are in the string.
String voltage higher than expected	Malfunction of one of the power optimizers	Test each optimizer output; replace malfunctioning unit.
String voltage higher than expected	SafeDC mode previously disabled using SolarEdge Key	Enable SafeDC using the SolarEdge Key.
Pairing to inverter failed	Power optimizers are shaded	Perform remote pairing via monitoring portal if connected.

Chapter 9: Technical Specifications

Single Phase Inverters

Parameter	Unit	SE2200	SE3000	SE3500	SE4000	SE5000	SE6000
Rated AC Power Output	W	2,200	3,000	3,500	4,000	5,000	6,000
AC Output Voltage (nominal)	Vac	220/230	220/230	220/230	220/230	220/230	220/230
AC Output Voltage Range	Vac	184–264.5	184–264.5	184–264.5	184–264.5	184–264.5	184–264.5
AC Frequency (nominal)	Hz	50/60 ±5	50/60 ±5	50/60 ±5	50/60 ±5	50/60 ±5	50/60 ±5
Max Continuous Output Current	A	12	16.5	19.5	22	27	27
RCD / RCD Step Detector	mA	300/30	300/30	300/30	300/30	300/30	300/30
Max Input Voltage	Vdc	500	500	500	500	500	500
Nominal DC Input Voltage	Vdc	350	350	350	350	350	350
Max Input Current	Adc	8.5	11.5	13.5	15.5	19.5	23
Max Inverter Efficiency	%	97.6	97.6	97.6	97.6	97.6	97.6
Euro Weighted Efficiency	%	97.6	97.6	97.5	97.5	97.4	97.4
Night-Time Power Consumption	W	<2.5	<2.5	<2.5	<2.5	<2.5	<2.5
Operating Temp Range	°C	-20 to +50	-20 to +50	-20 to +50	-20 to +50	-20 to +50	-20 to +50
Dimensions W×L×H	mm	540×315×172	540×315×172	540×315×172	540×315×172	540×315×191	540×315×191
Weight	kg	20.2	20.2	20.2	20.2	21.7	21.7
Cooling	–	Natural Conv.	Natural Conv.	Natural Conv.	Natural Conv.	Natural Conv.	Natural Conv.
Noise Level	dBA	<50	<50	<50	<50	<50	<50
Protection Rating	–	IP65	IP65	IP65	IP65	IP65	IP65

Parameter	Unit	SE2200	SE3000	SE3500	SE4000	SE5000	SE6000
DC Input Pairs	–	1 MC4	1 MC4	1 MC4	2 MC4	2 MC4	2 MC4
Max Altitude	m	2,000	2,000	2,000	2,000	2,000	2,000

Three Phase Inverters

Parameter	Unit	SE5K	SE7K	SE10K	SE12.5K	SE15K	SE17K
Rated AC Power Output	VA	5,000	7,000	10,000	12,500	15,000	17,000
AC Output Voltage L-L / L-N	Vac	380/220	380/220	380/220	380/220	380/220	380/220
AC Frequency	Hz	50/60 ±5	50/60 ±5	50/60 ±5	50/60 ±5	50/60 ±5	50/60 ±5
Max Continuous Output Current	A/ph	8	11.5	16	20	23	26
Max Input Voltage	Vdc	950	950	950	950	950	950
Nominal DC Input Voltage	Vdc	750	750	750	750	750	750
Max Input Current	Adc	8.5	12	16.5	21	22	23
Max DC Power (STC)	W	5,500	7,700	11,000	13,700	16,500	18,700
Max Inverter Efficiency	%	98	98	98	98	98	98
Euro Weighted Efficiency	%	97.3	97.3	97.6	97.7	97.6	97.7
Night-Time Power Consumption	W	<2.5	<2.5	<2.5	<2.5	<2.5	<2.5
Operating Temp Range	°C	-20 to +60	-20 to +60	-20 to +60	-20 to +60	-20 to +60	-20 to +60
Dimensions W×L×H	mm	540×315×260	540×315×260	540×315×260	540×315×260	540×315×260	540×315×260
Weight	kg	33.2	33.2	33.2	33.2	33.2	33.2
Cooling	–	Fan (user replaceable)	Fan	Fan	Fan	Fan	Fan
Protection Rating	–	IP65	IP65	IP65	IP65	IP65	IP65
Grid Connection	–	3/N/PE 230/400	3/N/PE 230/400	3/N/PE 230/400	3/N/PE 230/400	3/N/PE 230/400	3/N/PE 230/400

Parameter	Unit	SE5K	SE7K	SE10K	SE12.5K	SE15K	SE17K
Max Altitude	m	2,000	2,000	2,000	2,000	2,000	2,000

Standards Compliance

Safety	IEC-62103 (EN50178), IEC-62109
Grid Connection Standards	VDE-AR-N-4105, VDE 0126-1-1, AS-4777, RD-1663, DK 5940
Emissions	IEC61000-6-2, IEC61000-6-3, IEC61000-3-11, IEC61000-3-12, FCC part15 class B
Environmental	RoHS, WEEE compliant
Communication Interfaces	RS485, RS232, Ethernet, ZigBee (optional)
Protective Class	Class I
Overvoltage Category	III
Pollution Degree (inside/outside)	2/3

Recommended Circuit Breaker / Fuse Sizes

Inverter Model	Max Output Current (A)	Suggested Fuse Rating (A)
SE2200	12	16
SE3000	16.5	20
SE3500	19.5	25
SE4000	22	25
SE4000-16A	16	20
SE5000	27	32
SE6000	27	32
SE5k (3-phase)	8	10
SE7k (3-phase)	11.5	16
SE10k (3-phase)	16	20
SE12.5k (3-phase)	20	25
SE15k (3-phase)	23	25
SE17k (3-phase)	26	32

Chapter 10: Safe DC, Maintenance & Component Replacement

Appendix C: Safe DC

When AC supply to the inverter is shut off or when the inverter ON/OFF switch is turned to OFF, the DC voltage drops to a safe voltage of 1V per optimizer.

SolarEdge inverters are certified as disconnection devices for PV generators, compliant with:

- IEC 60947-3:1999 and amendments
- DIN EN 60947-3 / VDE 0660-107:2006-03
- IEC 60364-7-712:2002-05 / DIN VDE 0100-712:2006-06

Safe Disconnection Procedure

67. Turn the inverter ON/OFF switch to OFF, or disconnect AC by shutting off the AC breaker at the site. DC voltage displayed on LCD begins to decrease.
68. When DC voltage reaches a safe level, PV connectors at the inverter input can be disconnected. A galvanic separation then exists between the PV array and the inverter.

WARNING: Under single fault conditions, Safe DC voltage is only guaranteed when using modules up to: 95 Voc (single phase inverter) or 70 Voc (three phase inverter).

Appendix D: Fan Maintenance and Replacement

Three-phase inverters have two fans: Fan 1 (internal – requires SolarEdge technician) and Fan 2 (accessible from outside – user replaceable). A replacement kit is available from SolarEdge.

Fan Maintenance

69. Keep the fan and screen clean by blowing dust away when it accumulates, or at least once a year.
70. Check the Fan Health Status screen on the LCD.
71. Replace the fan if:

- The fan stops operating
- Fan status shows 'Replace'
- Fan status shows 'Not Working' (first try turning AC off and on)
- Error 'Fan 2 Failure' is displayed

Fan 2 Replacement Steps

72. Turn OFF the inverter ON/OFF switch and wait until LCD shows DC voltage is safe (<50V), or wait 5 minutes.

WARNING: If you cannot see the inverter panel or a malfunction is indicated, wait 5 minutes for capacitors to discharge.

73. Turn OFF the AC switch of the main distribution panel.
74. Use a Philips screwdriver to unfasten the single screw of the fan cover.
75. Open the fan door.
76. Disconnect the fan connector and remove the fan.
77. Connect the fan connector to the new fan.
78. Close the fan door and fasten the cover screw.
79. Check the Fan Health Status screen to confirm the new fan is operating.

Appendix E: Replacing and Adding System Components

NOTE: If permanently disassembling the installation or part of it, use disposal methods dictated by local regulations.

Replacing an Inverter

80. Turn OFF the inverter ON/OFF switch and wait until LCD shows DC voltage is safe (<50V), or wait 5 minutes.
81. Disconnect the AC by turning OFF circuit breakers on the distribution panel.
82. Open the inverter cover.
83. Disconnect the DC plugs and AC wires from the inverter.
84. Remove the two lower screws securing the inverter to the mounting bracket and remove the inverter.

NOTE: If you remove the old inverter and do not immediately install a new one, use insulation tape to isolate each AC and DC wire.

85. Place the new inverter on the mounting bracket; insert the screws.
86. Follow the instructions of Chapter 4 (Installing the Inverter) and Chapter 5 (Commissioning the Installation).

Adding, Removing or Replacing Power Optimizers

87. Turn OFF the inverter ON/OFF switch and wait until safe (<50V), or wait 5 minutes.
88. Turn OFF the AC switch of the main distribution panel.
89. Disconnect and connect the necessary power optimizers.
90. Perform pairing and commissioning procedures on all inverters affected.
91. In the monitoring portal, replace the serial number of the removed optimizer with the serial number of the newly installed optimizer.

Support and Contact Information

Before contacting support, have the following information ready:

- Inverter and power optimizer model numbers
- Serial number of the product in question
- The error indicated on the inverter screen or monitoring portal
- System configuration information (type and number of modules, number and length of strings)
- Communication method to the SolarEdge monitoring portal
- Inverter software version (visible in the ID status screen)

Region / Country	Phone	Email
US & Canada	1 877 360 5292	ussupport@solaredge.com
Belgium	080073041	support@solaredge.be
France	0800917410	support@solaredge.fr
Germany	+49 89-45459730	support@solaredge.de
Italy	800 784 824	support@solaredge.it
Japan	+81.3.5530.9360	support@solaredge.jp
Australia	1800149229	support@solaredge.com
Greece	00800125574	support@solaredge.com

Region / Country	Phone	Email
Israel	+972 73 240-3118	support@solaredge.com
Netherlands	08000221089	support@solaredge.com
United Kingdom	0800 028 1183	support@solaredge.com
APAC (Asia Pacific)	—	support-asia@solaredge.com
Worldwide	+972 73 240-3118	support@solaredge.com
Fax	+972 73 240-3117	—

NOTE: Website: www.solaredge.com | Support portal: http://www.solaredge.com/groups/support/services